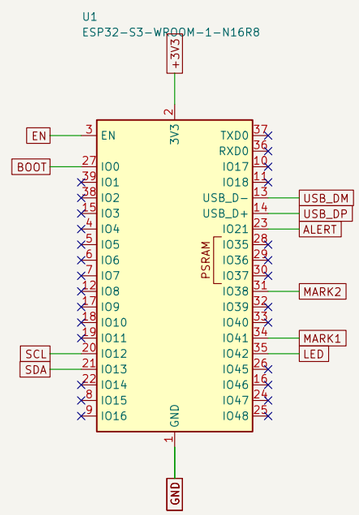
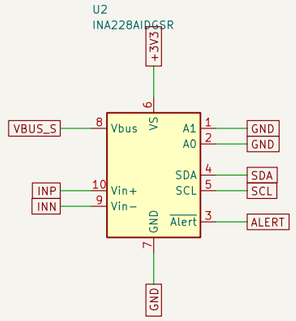
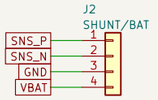
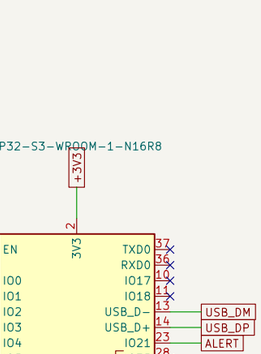

U1
ESP32-S3-WROOM-1-N16R8
+3V3
EN
BOOT
SCL
SDA
USB_DM
USB_DP
ALERT
MARK2
MARK1
LED
PSRAM
GND

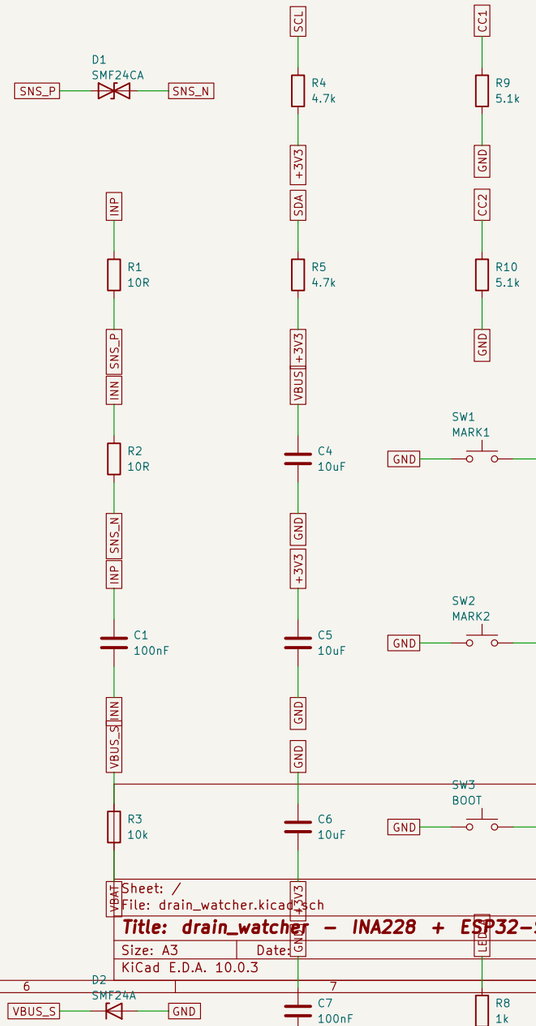

U2
INA228AIDGSR
+3V3
VBUS_S
Vbus
VS
A1
A0
GND
INP
INN
Vin+
Vin-
SDA
SCL
Alert
ALERT
GND


J2
SHUNT/BAT
SNS_P
SNS_N
GND
VBAT


J1
USB-C
VBUS
CC1
CC2
D-
D+
SBU1
SBU2
SHIELD
GND
SH


U3
AP2112K-3.3
VIN
VOUT
EN
GND
VBUS
+3V3

#FLG0
PWR_FLAG
GND
#FLG1
PWR_FLAG
VBUS


D1
SMF24CA
SNS_P
SNS_N
D2
SMF24A
VBUS_S
GND
R1
10R
R2
10R
R3
10k
C1
100nF
R4
4.7k
R5
4.7k
C4
10uF
C5
10uF
C6
10uF
C7
100nF
R9
5.1k
R10
5.1k
R8
1k
SW1
MARK1
SW2
MARK2
SW3
BOOT
SCL
SDA
INP
INN
VBUS
+3V3
CC1
CC2
LED
VBAT